

SUNIX SDC Expansion Board

gpiod

User Guideline

Driver Ver. 2.1.1.0

Doc Ver. 1.0

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1. gpiod

Confirm that the following packages have been installed:

➤ gpiod

Ubuntu/Debian

```
$apt-get install gpiod libgpiod-dev
```

1.1 Detect the SDC hardware

Make sure the SDC hardware device and SDC GPIO driver are installed

```
$sdc-config -d
```

```
root@test-System-Product-Name:~# sdc-config -d
ID0: model=SDC0880I, firmware=1.9, pci_bus=3, irq=16
```

```
$sdc-config -D {ID}
```

```
root@test-System-Product-Name:~# sdc-config -D 0
ID0: model=SDC0880I, firmware=1.9, pci_bus=3, irq=16
CONFIG(0):
  ver=x00, pid=1, /dev=sdc_config0
  brand=1, model=0
DIO(1):
  ver=x01, pid=2, /dev=gpiochip1
  cap=x03, nr_bank=2
    B0: nr_io=8, cap=input
    B1: nr_io=8, cap=output
root@test-System-Product-Name:~#
```

Check if GPIO Devices have SDC hardware.

```
$gpiodetect
```

```
root@test-System-Product-Name:~# gpiodetect
gpiochip0 [INTC1085:00] (473 lines)
gpiochip1 [sdc_gpio.2.auto] (16 lines)
root@test-System-Product-Name:~#
```

Digital IO channels list, this example is 8DI/8DO.

`$gpiointfo gpiochipn`

```
root@test-System-Product-Name:~# gpiointfo gpiochip1
gpiochip1 - 16 lines:
line 0:      unnamed      unused  input  active-high
line 1:      unnamed      unused  input  active-high
line 2:      unnamed      unused  input  active-high
line 3:      unnamed      unused  input  active-high
line 4:      unnamed      unused  input  active-high
line 5:      unnamed      unused  input  active-high
line 6:      unnamed      unused  input  active-high
line 7:      unnamed      unused  input  active-high
line 8:      unnamed      unused  output  active-high
line 9:      unnamed      unused  output  active-high
line 10:     unnamed      unused  output  active-high
line 11:     unnamed      unused  output  active-high
line 12:     unnamed      unused  output  active-high
line 13:     unnamed      unused  output  active-high
line 14:     unnamed      unused  output  active-high
line 15:     unnamed      unused  output  active-high
root@test-System-Product-Name:~#
```

0 ~ 7 are DI

8 ~ 15 are DO

1.2 Digital IO Control

The example hardware setting is DO1 connect to DI1.

Set output value

`$gpioset gpiochipn {linex(must be output)} = {value}`

```
root@test-System-Product-Name:~# gpioset gpiochip1 8=1
root@test-System-Product-Name:~# gpioset gpiochip1 8=0
root@test-System-Product-Name:~#
```

DO1 Pull high

DO1 Pull low

Get input value

`$gpioget gpiochipn {linex(must be input)}`

```
root@test-System-Product-Name:~# gpioget gpiochip1 0
1
root@test-System-Product-Name:~# gpioget gpiochip1 0
0
```

DI1 is high

DI1 is low

Get input event

```
$gpiomon --num-events=5 gpiochipn {linex(must be input)}
```

```
--num-events=5 : Captures 5 events before exiting.
```

```
root@test-System-Product-Name:~# gpiomon --num-events=5 gpiochip1 0
event:  RISING EDGE offset: 0 timestamp: [    462.322665142] → DI1 rising event
event:  FALLING EDGE offset: 0 timestamp: [    464.605233685] → DI1 falling event
□
```